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Building Security Into Your Palm™ Applications



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3Com
More connected.

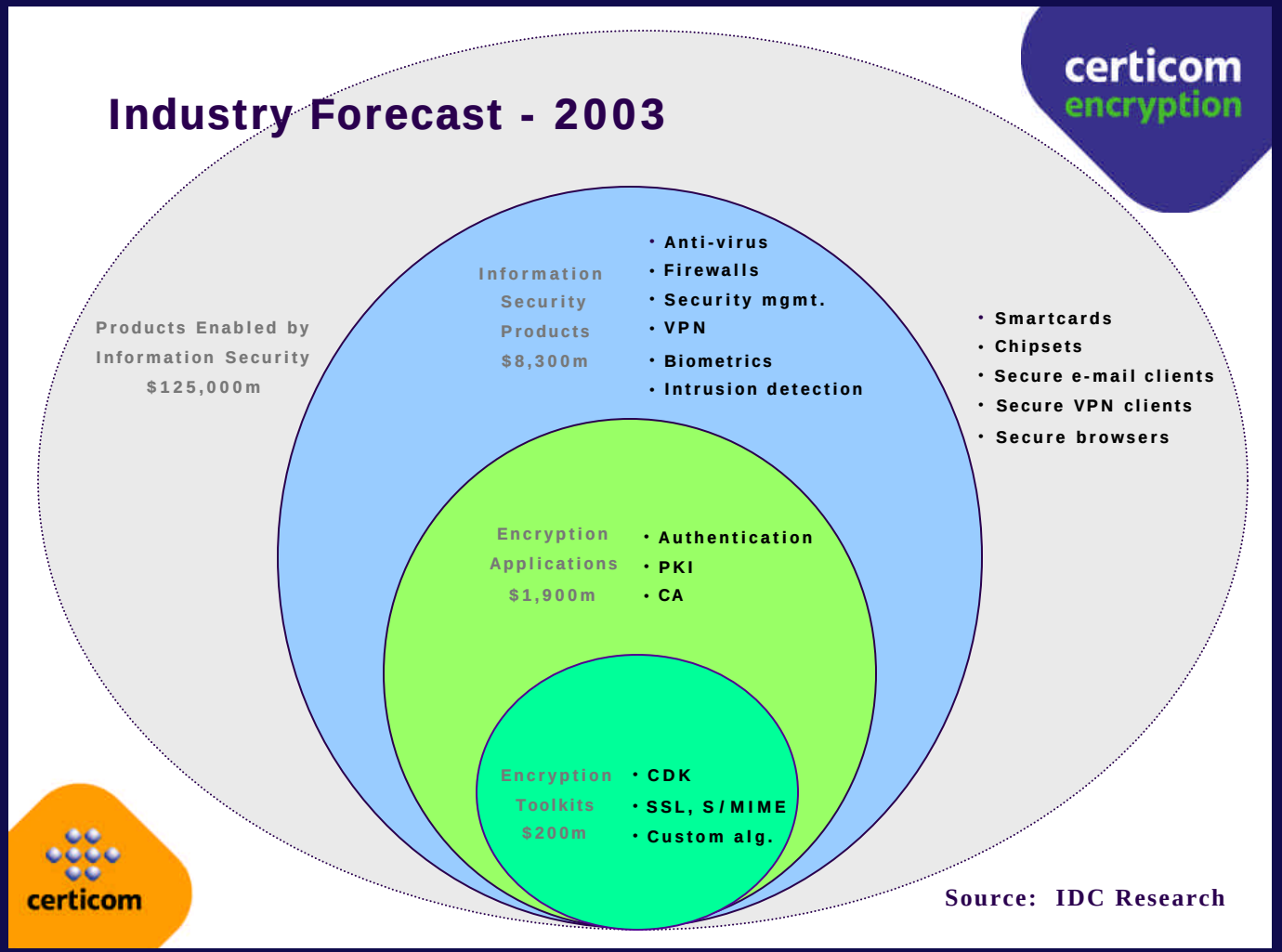
Information Security Overview

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- Packaged Protection
  - Cryptography
  - Toolkits and products
  - Services
- e-Business Enabler
  - Electronic commerce
  - Mobile enterprise workers
- $8.3b Industry in Year 2003

Information Security Industry

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Crypto 101

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- Security Services
  - Confidentiality
  - Authentication
  - Data integrity
  - Non-repudiation
- Symmetric-key Cryptography
  - DES
- Public-key Cryptography
  - RSA (factorization problem)
  - ECC (discrete logarithm problem)

Crypto 101 Continued

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- Internet Security Protocols
  - SSL
  - WTLS
  - IPSEC

Typical Threats

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- Breaking and Entering
- Internal Attacks
- Credit Card Fraud
- Data Theft

Information Security Threats

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Breaking and Entering

- Problem
  - Internet hackers attack your network
  - 23% of companies reported it last year
- Impact
  - Business operations setback
  - Loss of reputation
  - Financial
- Solution
  - Firewalls
  - Remote access servers
  - Two-factor authentication

Information Security Threats

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Internal Attacks

- Problem
  - Employee access abuse
  - 52% of companies reported it last year
- Impact
  - Business operations setback
  - Loss of reputation
  - Financial
- Solution
  - Embedded application security
  - PKI/CA
  - Authorization servers and administration software

Information Security Threats

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Fraud

- Problem
  - Identity theft
  - Credit card fraud
- Impact
  - Slows adoption of e-Commerce
  - Financial: banks, merchants, and individuals share the pain
- Solution
  - Secure communication links
  - Certificate services

Information Security Threats

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Data Theft

- Problem
  - Lost and stolen computing equipment
- Impact
  - Business operations setback
  - Loss of reputation
  - Financial
- Solution
  - Device-level file and data encryption
  - Secure applications

Information Security Products Used

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- Top Selling Security Product Categories
  - Virus protection
  - Firewalls
  - Access controls
  - Security management and administration
  - Intrusion detection
  - PKI/CA
  - File encryption
  - Encryption toolkits

Palm Computing Platform: The Big Enterprise Security Issue

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- Current Enterprise Policy
  - PDAs are not allowed on corporate networks
- Evidence: CSI Newsletter Excerpts
  - “An exception approved by Information Security is required to attach a PDA to company systems.”
  - “PDAs cannot be connected directly to the network, only to an approved host workstation.”
- Evidence: Gartner Group
  - “The vast majority of large businesses do not yet deploy company software to the devices.”
  - “Large firms are wasting resources by refusing to integrate employee-owned devices into their corporate computing infrastructure.”

Palm Computing Platform: Specific Security Issues

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- Secure Wireless Links for e-Commerce
  - Internet appliances (Palm VII™)
  - Financial services requirements (smart cards)
- Enterprise Data Synchronization
- Device and Data Theft
- Code and Data Exchange
  - Beaming
  - Viruses

Why Solve the Problem? Tremendous Opportunity

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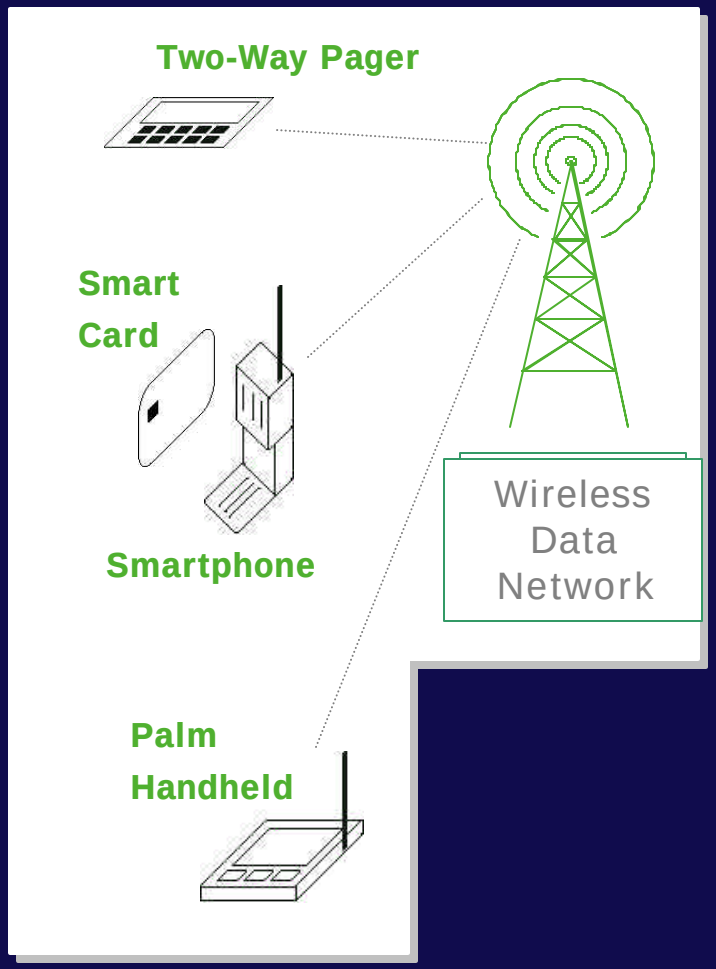
- Wireless Networks
- Enterprise Productivity
- Consumer Electronic Commerce
- Financial Services

Implementing A Solution

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- Give Your Potential Adversary a Challenge
- Build In Security From the Beginning
- Get it Right; Work with a Security Partner

Specific Solutions: Wireless Networks



- e-Commerce Enabler
- Early Examples
  - Palm VII, Aether Systems, JP Systems, 724 Solutions, Actual
- Solution Options
  - Wireless (OPP, WTLS)
  - SSL
  - Digital signatures

Specific Solutions: Wireless Financial Services

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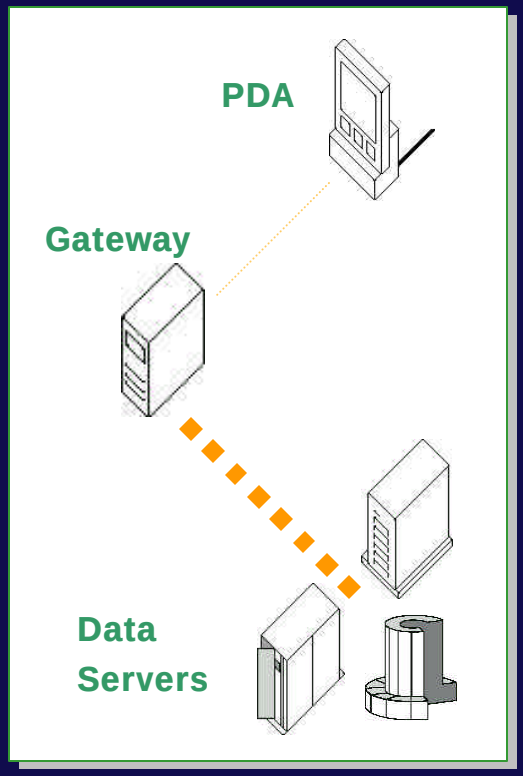
- Home Banking and Stock Trading
- Early Example
  - Newcom Funge
- Solution Options
  - Smart cards
  - Digital signatures
  - Embedded PKI



Specific Solution: Enterprise Productivity

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- One-button Sync of Enterprise Data
- Early Examples
  - Data synchronization gateways (AvantGo, Puma, Riverbed, ASL)
- Solution Options
  - SSL
- MAL Association



Specific Solution: Identity

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- Authenticated Clients
  - Tokens: strong authentication
  - Wallets: greater convenience
- Early Efforts
  - Many token attempts (Defender, e.ID, MicroToken, SecurID)
  - A wallet (ActivCard)
- Solution Options

Specific Solution: Data Theft

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- File and Data Encryption
- Early Efforts
  - Secure MemoPad
- Solution Options
  - Symmetric key
  - Public key

Putting the E in E-Business

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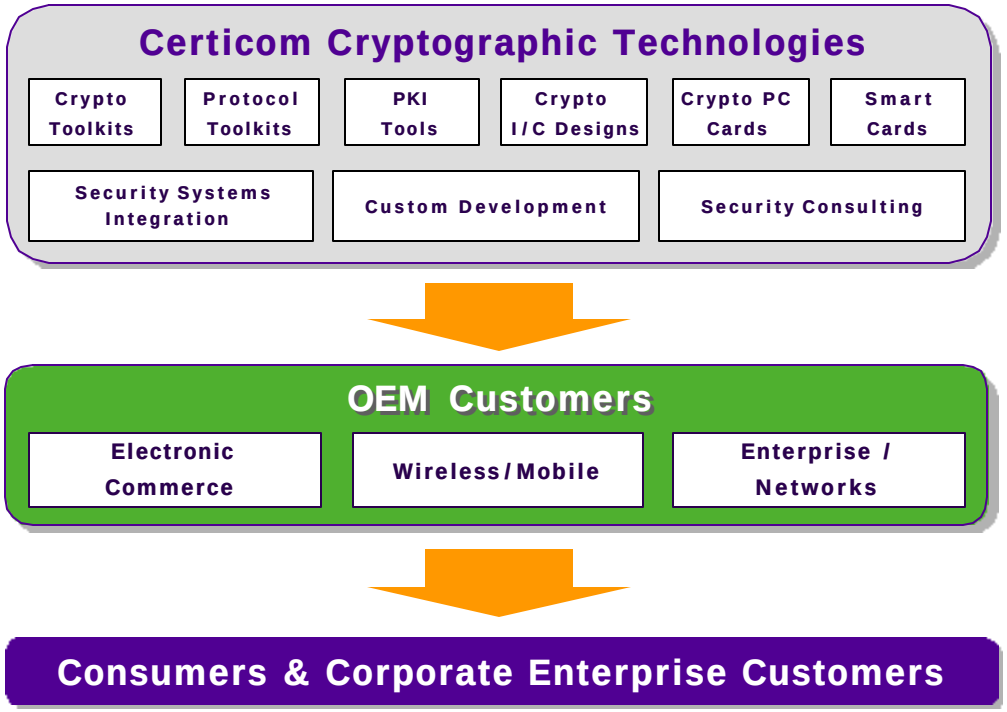
- Efficient Embedded Encryption
  - Open standards
  - High performance
  - Service and support
- Customers want to buy solutions
  - Security is a check box item
  - Look at home and car doors

Build It In

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Embedded Encryption Model

certicom



Toolkits for Open Standards

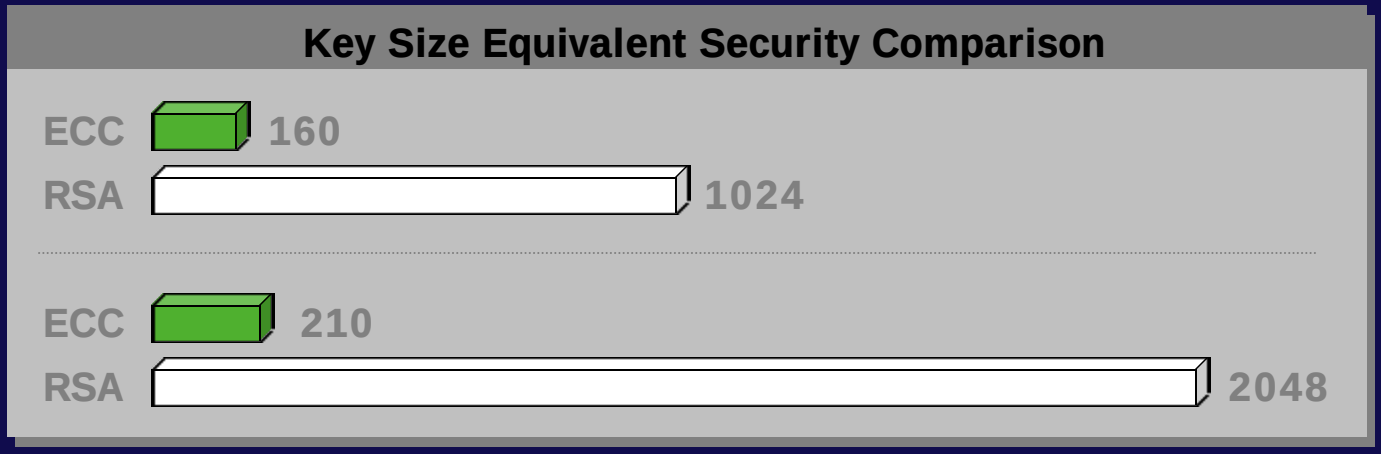
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- Standard Security Protocols
  - SSL
  - WTLS
  - IPSEC
- Certicom Toolkits
  - SSL Plus Family
    - Platforms: handheld/embedded, enterprise
    - Algorithms: ECC, DSA/DH, RSA
    - Languages: C and Java
  - WTLS Plus

Efficient Encryption

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- Elliptic Curve Cryptography
  - Public-key cryptography
  - Smaller keys with high cryptographic strength
  - Requires less processing power
  - Uses less network bandwidth
  - Reduces storage requirements



Certicom Brand

- End-user Target
- Third-party Products
- Meaning
  - Trust
  - Safe products
- Co-marketing Opportunities



Specific Brand Usage

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Palm VII™ Wireless Connected Organizer



Summary

- Information Security Requirement
- Huge e-Business / e-Commerce Opportunity
- Palm Computing Platform Security Solutions
- Your General Contractor for Security




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